1.

(a) (Application of Theory)

12.5MHz

(b) (Application of Theory)

$$\frac{5679 \cdot 10^{12}}{s^2 + 37{,}68 \cdot 10^6 \cdot s + 5679 \cdot 10^{12}}$$

(c) (Application of Theory)

68.6%

(d) (Application of Theory)

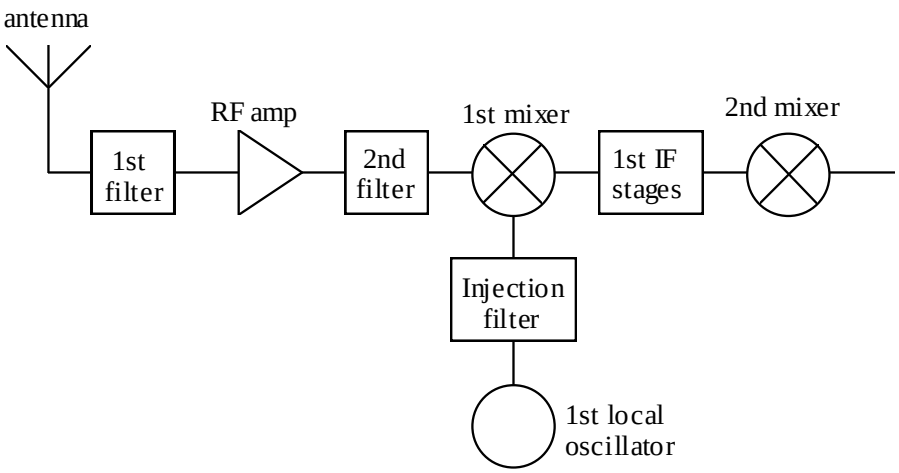
8.6MHz to 9.1MHz

(e) (Theory)

High gain reduces the effects of noise of subsequent stages, but it also can potentially affect linearity.

(f) (Theory and Application of Theory)

It is a second order low pass filter. Low pass filters are useful as explained in the diagram below:

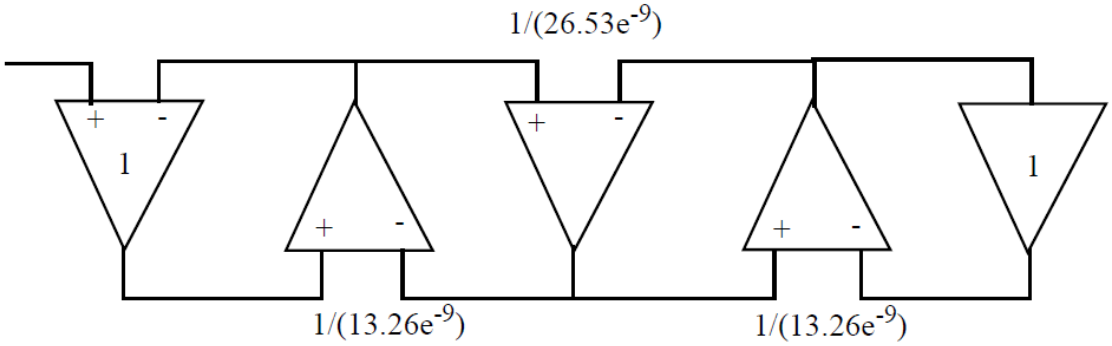


• 1st filter (preselector): Limits bandwidth of spectrum reaching the RF amplifier (to reduce IM distortion); attenuates receiver spurious responses (esp. image frequency); attenuates oscillator and first IF re-radiation which may be picked up by the antenna.

• 2nd filter: Attenuates noise at the image frequency, and second harmonics which may originate in the RF amplifier.

• Injection filter: Attenuates wideband noise around the LO frequency, and attenuates oscillator harmonics.

(g) (Application of Theory)



2.

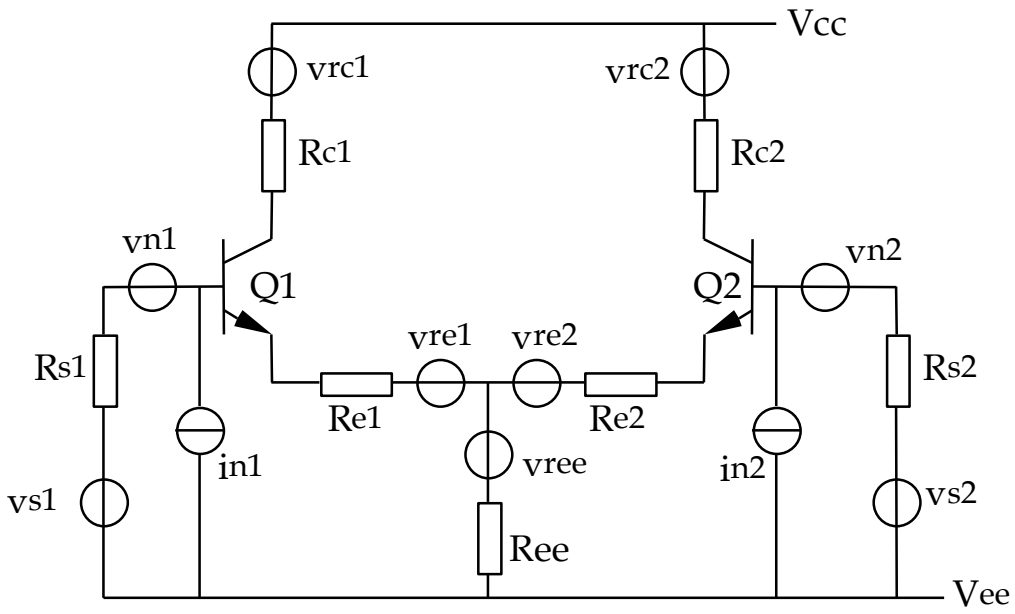
(a) (Theory)

Because a differential configuration gets rid of even order harmonics, hence reducing distortion.

(b) (Theory)

By using degeneration it is possible to increase the linearity of the differential pair. The drawback is that the efficacy of this technique relies on the value of the resistor (i.e the larger the resistor the better the linearity), but this also compromises noise and control over the transconductance.

(c) (Application of theory)



Note that this answer considers resistive input sources, but an answer would also be valid if Rs1 and Rs2 are considered 0.

- Thermal noise for resistors
- Thermal, Flicker and shot noise for transistors.

(d) (Application of theory)

The thermal noise of the degeneration resistors and the input noise of the transistors (assuming the biasing resistors are large). This is because the gain from those nodes to the input is low, so the noise "appears" reflected back. The noise from the biasing resistor is common mode so it has no effect.

(e) (Application of theory)

The advantage of removing the resistors is lower noise and controllability of the transconductance with the biasing current. The disadvantage is the loss of linearity.

(f) (Application of Theory)

276.5μV

(g) (Application of Theory)

Put the block with the lowest gain the last one in the chain.

3.

(a) (Application of theory)

By changing the sizes of the two input transistors the transconductance function's shape remains but it appears shifted with respect to the input in the x axis. The operating region of the transconductor is where the gradient of the transconductance function is the lowest (i.e. in the centre). By shifting the curve in the x-axis the optimum operating point is shifted. If the transconductance function of two differential pairs is shifted in opposite directions in the x-axis, but still kept close to each other, the linear regions add up with a small dip in the middle (which can be controlled with the sizes).

(b) (Application of theory)

They couldn't be the same because if they were the output of the circuit would be zero. By making it different it is possible to potentially cancel out high order harmonics, hence increasing linearity.

(c) (Application of theory)

For a single transistor:

$$\mathrm{Id} = \mathrm{Ido} + \left(\frac{\partial \mathrm{Ic}}{\partial \mathrm{Vin}}\right)\mathrm{Vin} + \left(\frac{\partial^2 \mathrm{Id}}{\partial \mathrm{Vin}^2}\right)\frac{\mathrm{Vin}^2}{2} + \left(\frac{\partial^3 \mathrm{Id}}{\partial \mathrm{Vin}^3}\right)\frac{\mathrm{Vin}^3}{6} + \ldots\ldots$$

where the derivatives are evaluated at the quiescent point Id = Ido

For the circuit above:

$$\mathrm{Iout} = (\mathrm{Id1} - \mathrm{Id2}) - (\mathrm{Id3} - \mathrm{Id4})$$

Thus $$\mathrm{Iout} = \frac{\mathrm{Vd}}{2}\left(\left(\frac{\partial \mathrm{Id}}{\partial \mathrm{Vd}}\right)_1 + \left(\frac{\partial \mathrm{Id}}{\partial \mathrm{Vd}}\right)_2 - \left(\frac{\partial \mathrm{Id}}{\partial \mathrm{Vd}}\right)_3 - \left(\frac{\partial \mathrm{Id}}{\partial \mathrm{Vd}}\right)_4\right)$$

$$+ \quad \frac{\mathrm{Vd}}{48}\left(\left(\frac{\partial^3 \mathrm{Id}}{\partial \mathrm{Vd}_3}\right)_1 + \left(\frac{\partial^3 \mathrm{Id}}{\partial \mathrm{Vd}^3}\right)_2 - \left(\frac{\partial^3 \mathrm{Id}}{\partial \mathrm{Vd}^3}\right)_3 - \left(\frac{\partial^3 \mathrm{Id}}{\partial \mathrm{Vd}^3}\right)_4\right)$$

where $\left(\frac{\partial^n \mathrm{Id}}{\partial \mathrm{Vd}^n}\right)_m$ is the nth derivative of the mth transistor

Since M1 and M2 are matched, and M3 and M4 are matched, then

$$\left(\frac{\partial^n \mathrm{Id}}{\partial \mathrm{Vd}^n}\right)_1 = \left(\frac{\partial^n \mathrm{Id}}{\partial \mathrm{Vd}^n}\right)_2 \text{ and } \left(\frac{\partial^n \mathrm{Id}}{\partial \mathrm{Vd}^n}\right)_3 = \left(\frac{\partial^n \mathrm{Id}}{\partial \mathrm{Vd}^n}\right)_4$$

To eliminate the third harmonic term, the transistors should be scaled appropriately so that $\left(\frac{\partial^3 \mathrm{Id}}{\partial \mathrm{Vd}^3}\right)_1 = \left(\frac{\partial^3 \mathrm{Id}}{\partial \mathrm{Vd}^3}\right)_3$ however we must ensure that $\left(\frac{\partial \mathrm{Id}}{\partial \mathrm{Vd}}\right)_1$ and $\left(\frac{\partial \mathrm{Id}}{\partial \mathrm{Vd}}\right)_3$ don't also cancel.

First order derivative $\frac{\partial \mathrm{Id}}{\partial \mathrm{Vd}} = \sqrt{\beta \mathrm{Is}}$ Third order derivative $\frac{\partial^3 \mathrm{Id}}{\partial \mathrm{Vd}^3} = \sqrt{\beta^3 / 8\mathrm{Is}}$

For cancellation of the third order coefficients:

$$(\beta 1/\beta 2)^{3/2} = \sqrt{(\mathrm{Is1}/\mathrm{Is2})} \text{ thus } \frac{(\mathrm{W1}/\mathrm{L1})^3}{(\mathrm{W2}/\mathrm{L2})^3} = \frac{\mathrm{Is1}}{\mathrm{Is2}}$$

(Note that if $\beta 1 = \beta 2$ and Is1 = Is2, the first order terms will also cancel!)

The total transconductance will then be:

$$gm = \sqrt{Is\beta_1} - \sqrt{Is\beta_2}$$

(d) (Application of theory)

The third order harmonic wouldn't be eliminated. This would translate to a compromised linearity.

(e) (Application of theory)

The transistors are mismatched

4.

(a) (New theory)

The current is ~0

(b) (New theory)

She realizes that the feedback to keep the drain of the input transistor at a fixed voltage is not properly implemented. Instead of negative feedback it is positive feedback, which makes the circuit saturate.

(c) (New theory)

The input transistor is likely to have left the ohmic region.

M1 needs to be in ohmic, M2 and M3 in saturation.

(d) (New theory)

$$Iout = \beta 1 \cdot (Vin - V_T)Vb$$

(e) (New theory)

$$Gm = \beta 1 \cdot V_b$$